INCH-POUND

MIL-D-85040B
20 May 1989
SUPERSEDING
MIL-D-85040A
10 November 1983

MILITARY SPECIFICATION

DRAWERS AND UNDERSHIRTS, FLYER'S, ANTI-EXPOSURE, ARAMID, HIGH TEMPERATURE RESISTANT, CWU-43/P AND CWU-44/P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type and five sizes of a high temperature resistant, aramid, anti-exposure, flyer's drawers and undershirts designated CWU-43/P for the drawers and CWU-44/P for the undershirts.

1.2 Classification. The CWU-43/P drawers and the CWU-44/P undershirts, as applicable, shall be furnished in the following sizes, as specified (see 6.2(c)):

SIZE				
X-SMALL	SMALL	MEDIUM	LARGE	X-LARGE

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

JJ-W-155 Webbing, Textile, (Cotton, Elastic)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department, (SESD), (Code 53) Lakehurst, NJ, 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC - N/A FSC 8415

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UU-P-268	Paper, Kraft, Untreated, Wrapping
DDD-L-20	Label, For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

MILITARY

MIL-T-83193	Thread, Aramid Spun, Staple
MIL-C-85038	Cloth, Knitted, Aramid, High Temperature Resistant

STANDARDS

FEDERAL

FED-STD-751	Stiches, Seams, and Stitching

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129	Marking for Shipment and Storage

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials and components. The materials and components shall conform to the applicable specifications, purchase description, and standards as listed or required herein.

3.2.1 Cloth.

3.2.1.1 Body. The body of the drawers and the undershirts shall be fabricated from the aramid, high temperature resistant cloth conforming to MIL-C-85038, Type I. The color shall match the standard sample (see 6.3).

3.2.1.2 Cuffs. The cuffs, for the drawers and the undershirts, shall be fabricated from the aramid, high temperature resistant cloth conforming to MIL-C-85038, Type II. The color shall match the standard sample (see 6.3).

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3.2.1.3 Collarette. The collarette (mock turtleneck), for the undershirts, shall be fabricated from the aramid, high temperature resistant cloth conforming to MIL-C-85038, Type II. The color shall match the standard sample (see 6.3).

3.2.1.4 Binding and reinforcement. The binding and reinforcement, for the fly area of the drawers and for the collarette and shoulder seam of the undershirt, shall be fabricated from flat high temperature resistant aramid jersey. The cut width of the binding and reinforcement shall be sufficient to provide for a finished width of 1/2 ± 1/8 inch.

3.2.2 Elastic webbing.

3.2.2.1 Waistband. The elastic webbing, for the waistband, shall conform to JJ-W-155, Type I, Class 9, color unbleached. The cut lengths, for the various sizes, shall be compatible with 3.5 and Table III.

3.2.2.2 Stirrups. The elastic webbing, for the stirrups, shall conform to JJ-W-155, Type I, Class 3, 3/4 inch wide, color optional. The cut lengths, for the various sizes, shall be a nominal 13 inches (see Figure 9).

3.2.3 Sewing Thread. The sewing thread for the various seams and stitch types shall conform to MIL-T-83193. The color shall match the standard sample(see 6.3.1).

3.3 Design. The CWU-43/P drawers and the CWU-44/P undershirts shall conform in appearance to Figures 1 through 5, as applicable.

3.4 Construction. The construction of the drawers and undershirts, as applicable, shall conform to Figures 1 through 12.

3.4.1 Cutting. The drawers and undershirts, as applicable, shall be cut in strict accordance with the patterns furnished by the Government (see 6.3) which show direction lines for cutting and notches or markings, for the proper assembly of all the parts. The Government patterns shall not be altered in any manner and shall be used for making the work patterns. The working patterns shall be identical to the Government furnished patterns. The wales of the body and the cuffs shall be in the length (vertical) direction. The wales of the mock turtle collarette shall be, in the direction, perpendicular to the seam.

NOTE

Due to the inherent stretch of the cloth, it is recommended that cutting lays be limited to a maximum of 48 plies for the purpose of ensuring proper component dimensions.

3.4.1.1 List of pattern parts. The pattern parts required for the manufacture of the drawers and undershirts, as applicable, are as follows:

CWU-43/P Drawers	CWU-44 Undershirts
Front Fly Panel	Sleeve
Back Crotch Panel	Collarette
Leg Gusset	Cuff
Cuff	Body Section
Leg Section	

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3.4.2 Size marking. The component parts of the drawers or undershirts, as applicable, shall be marked or ticketed, by any commercial method, for the proper assembly of the drawers or undershirts, as applicable. No metal fastening device or sewn-on ticket shall be used. The size markings shall be accomplished by the use of an ink pad numbering machine, rubber stamp, or pencil, provided the markings do not show through to the outside of the drawers or undershirts, as applicable, and are not deleterious to the cloth. The markings, which are placed on the seam allowance, shall not be visible on the outerside. The adhesive type size marking tickets may be used, for the size markings, provided, they conform to 3.4.2.1. The assembled drawers or undershirts, as applicable, shall not contain any size marking tickets.

3.4.2.1 Adhesive size marking tickets. The adhesive size marking tickets shall be fabricated from paper with a thermo-activated adhesive appplied on one side. The adhesive shall not discolor or damage the cloth and the adhesive mass shall not adhere to the cloth, upon removal of the ticket. The heat used to attach the thermo-activated adhesive tickets shall not stiffen, harden, scorch, or damage the cloth in any manner.

3.4.3 Seams and stitching. All the seams and stitching used in the fabrication of the drawers and the undershirts shall conform to Figures 1 through 12, as applicable. The seam and stitch types, specified in Figures 6 through 12, shall conform to FED-STD-751. The width of the stitches and the stitches per inch, for the various types of the stitching, shall conform to Table I. Unless otherwise specified, width of seam stitching shall be ±1/16 inch.

3.4.3.1 Sewing. Each row of stitching shall be straight and parallel to the seam edge. The straightness of the stitching, in any row, shall be maintained within a tolerance of plus or minus 1/32 inch. The ends of the Stitch Type 301 stitching shall be backstitched, by overlapping on themselves, a minium of 1/4 inch. Thread breaks, skips, and run-off shall be overstitched with Stitch Type 301 stitching not less than one inch. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together. No seam or component shall be twisted, puckered, or pleated and no part of the drawers or the undershirts, as applicable, shall be caught in an unrelated operation or seam. All the thread ends, or threads beginning or terminating a row of stitching, shall be trimmed to a maximum length of 3/8 inch. No raw or cut cloth edge shall be visible. All the seam edges shall be properly forced out and shall not contain any folds.

3.4.3.2 Bartacking.

3.4.3.2.1 Cuff Seams. A 1/4 inch bartack shall be positioned and centered across each cuff seam see (Figures 1 and 4.). Each bartack shall contain a minimum of 14 stitches.

3.4.3.2.2 Stirrup Attachments. Three 1/2 inch bartacks may be used on the stirrup attachments in lieu of a crossboxstitch (see Figure 9.) Each bartack shall contain a minimum of 26 stitches.

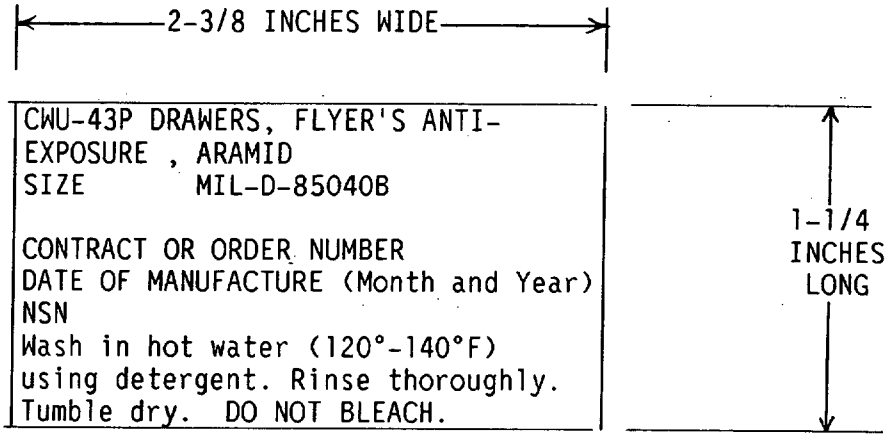
3.5 Dimensions. The dimensions, for the various sizes of the drawers and undershirts, as applicable, shall conform to Table II, when determined as specified in 4.5.2 through 4.5.2.2 and Figures 13 and 14, as applicable. The

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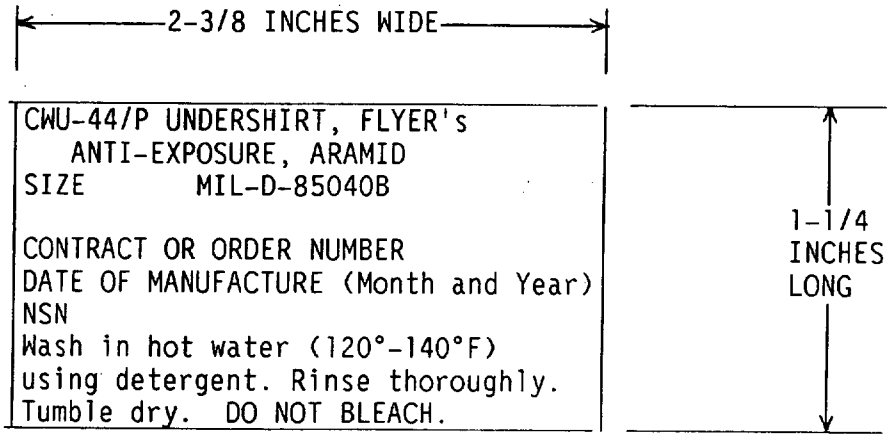
variation in length between the left and right legs, or left and right sleeves shall not exceed 1/2 inch in any drawers or undershirt, as applicable.

3.6 Label. The identification label, for the drawers and the undershirts, as applicable, shall conform to DDD-L-20, Type VI, Class 1, Launderable Label, except that the item description shall be 8 point and all the other information shall be 6 point.

3.6.1 Drawers. The identification label, for the drawers, shall be centered, within 1/2 inch, and sewn to the inner back portion, of the waistband, as specified in Figure 8, in a manner that shall permit the label to expand with the waistband of the drawers. The top edge of the label shall be aligned with the top edge of the waistband. The label shall be attached on the sides only (vertical stitching) when the 2 sided stitch is employed or around the label perimeter when the 4 sided stitch method is used. The label size and the information, for the drawers, shall be as follows:



3.6.2 Undershirt. The identification label, for the undershirt, shall be centered, within 1/2 inch, and sewn to the back, on the innerside of the undershirt, 1/4 inch below the collarette seam reinforcement, as specified in Figures 5 and 11. The label shall be attached on the sides only (vertical stitching) when the 2 sided stitch method is employed and around the label perimeter when the 4 side stitch method is used. Labels shall be attached in such a manner that shall permit the label to expand with the fabric of the undershirt. The label size and the information, for the undershirts shall be as follows:



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3.7 Workmanship. After completion of the final assembly, the drawers or undershirts, as applicable, shall be thoroughly cleaned and all thread scraps, lint, and foreign matter shall be removed. The drawers or undershirts, as applicable, shall not contain any tear, cut, mend, needle chew, abraded area, burn, spot, stain, or hole. The drawers or undershirts, as applicable, shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, or durability. The drawers or undershirts, as applicable shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Resposibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept of defective material.

4.2 Classification of inspection. The examination and testing of the high temperature resistant, aramid, anti-exposure, flyer's, CWU-43/P drawers or CWU-44/P undershirts, as applicable, shall be classified as follows:

a. First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 3.1 and 4.3 through 4.3.2).

b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.4 through 4.4.1.2).

4.3 First article inspection. The first article inspection of the high temperature resistant, aramid, anti-exposure, flyer's, CWU-43/P drawers or CUW-44/P undershirts, as applicable, shall consist of examinations and tests for all of the requirements of this specification.

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4.3.1 First article samples. Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit two pairs of drawers of undershirts, as applicable, of any size specified in the contract or order. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these pairs of drawers or undershirts, as applicable, from contract to contract, submission of further first article inspection samples, on the new contract, may be waived at the discretion of the procuring activity (see 6.2(d)). Approval of the first article inspection samples or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2(e)).

4.3.2 Upon completion of the first article inspection, all the applicable inspection reports and when aplicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the cognizant Government activity. One of the approved first article inspection samples of the drawers or undershirts, as applicable, will be returned to the manufacturer for use in monitoring production. The other drawers or undershirt, as applicable, will be destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under the contract or order.

4.4 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:

Visual examination of the drawers or undershirts, as applicable
Dimensional check of the drawers or undershirts, as applicable
Packaging

4.4.1 Sampling.

4.4.1.1 Inspection lot.

4.4.1.1.1 Drawers. An inspection lot size shall be expressed in units of one pair of drawers typical of production garment quality. The sample unit shall be one pair of drawers.

4.4.1.1.2 Undershirts. An inspection lot size shall be expressed in units of one undershirt typical of production garment quality. The sample unit shall be one undershirt.

4.4.1.1.3 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing pairs of drawers or undershirts, as applicable, of one size. The drawers or undershirts, as applicable, shall be typical of production garment quality and be fully prepared for delivery. The sample unit shall be one shipping container, containing pairs of drawers or undershirts, as applicable, of one size, fully prepared for delivery with the exception that it need not be sealed.

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4.4.1.2 Sampling for the examinations and tests of the drawers, undershirts, and packaging. The sample size, acceptance criteria, examinations, and tests required, for the drawers, undershirts, and the packaging as applicable, shall be as specified in Table III.

4.5 Inspection methods.

4.5.1 Visual examination.

4.5.1.1 Drawers and undershirts. Each pair of drawers or each undershirt, as applicable, selected as a sample unit from the lot, shall be thoroughly checked dimensionally and examined visually to determine conformance to this specification. The classification and list of defects, Tables IV and V, as applicable, shall be used to classify and enumerate the defects found.

4.5.1.2 Packaging. Each of the fully prepared shipping containers, containing pairs of drawers or undershirts, as applicable, of one size, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects, Table VI, shall be used to enumerate the defects.

4.5.2 Procedure for measurement. The drawers of the undershirt, as applicable, to be measured shall be laid flat, face up, without tension, on a smooth, flat surface so that any creases or wrinkles shall not affect the measurement.

4.5.2.1 Drawers. The drawers shall be measured in accordance with Figure 13, as follows:

a. Waist. Measure the width of the waist along the top edge of the relaxed elastic waistband.

b. Back rise. Measure the back rise, along the seat, from the top of the waistband, at the back, to the bottom folded edge at the crotch.

c. Length. Measure the length of each leg in a straight line, from the top edge of the waistband at the height of the center back of waist to the bottom edge of the cuff along the outer side of the leg.

d. Calf. Measure the width of each calf, at a point 10 inches up from the cuff attaching seam, from side fold to side fold.

e. Fly opening. Measure the length of the fly opening.

f. Cuff length. Measure the length of each cuff, from the attaching seam to the end of the cuff.

g. Cuff width. Measure the width of the cuff, at the unattached end, from side fold to side fold.

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4.5.2.2 Undershirts. The undershirts shall be measured in accordance with Figure 14 as follows:

h. Body length. Measure the body length on the front, from the outer side edge of the collarette, at the shoulder, to the bottom edge of the undershirt.

i. Body width. Measure the body width, one inch below the armhole seam, from side fold to side fold, or from side seam to side seam, whichever is applicable.

j. Armhole. Measure the width of the armhole, from the sleeve inseam to the fold at the shoulder.

k. Neck opening. Measure the width of the neck opening at the collarette attaching seam from the side fold to the side fold of the collarette.

l. Sleeve inseam. Measure the length of each sleeve inseam, from the armhole seam to the bottom of the cuff.

m. Shoulder width. Measure the distance, on each shoulder, from the collarette attaching seam to the armhole seam.

n. Cuff length. Measure the length of each cuff, from the attaching seam to the end of the cuff.

o. Cuff width. Measure the width of the cuff, at the unattached end, from side fold to side fold.

p. Mock turtleneck length. Measure the length of the mock turtleneck (collarette), from the top fold to the attaching seam.

5. PACKAGING

5.1 Packaging. Packaging shall be Level A or C, as specified (see 6.2(f)).

5.1.1 Level A. Each pair of the drawers or undershirt, as applicable, shall be neatly folded,in accordance with the commercial practice, to an approximate size, 11-1/2 by 14-1/2 inches. A paper label shall be inserted (see 5.3.1) and each folded drawers or undershirt, as applicable, shall be sealed within a clear polyethylene bag of 0.0125 ±. 00025 inch film. The heat sealed seams shall be straight, continuous, and parallel to each other and the edgesof the polyethylene bag. The closing heat sealed seam of the bag shall be as close as possible to the edge of the bag. No drawers or undershirt, as applicable, shall be deformed, distorted, or bent in the bagged condition nor shall the polyethylene bag be damaged. No part of any drawers, or undershirt, as applicable, shall be caught in any heat sealed seam. One corner of each bag shall contain a hole, 1/4 inch in diameter, to allow the excess air to escape. As an alternate, the final closure of the bag may be accomplished by use of a tuck or reverse flaps. The approximate flat size of the polyethylene bag shall be 13 by 17 inches.

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5.1.2 Level C. The drawers or undershirts, as applicable, of one size, shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2(f)). Each shipping container shall contain only pairs of drawers or undershirts, as applicable of one size.

5.2.1 Level A. Fifty pairs of drawers or forty undershirts, as applicable, packaged as specified in 5.1.1, shall be packed as specified in 5.2.2, except that the fiberboard container shall be Weather Resistant Class, Variety SE, Grade V3c or V3s. In addition, each container shall be reinforced with flat steel strapping or tape banding in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Fifty pairs of drawers or forty undershirts, as applicable, packaged as specified in 5.1.1, shall be packed within a snug fitting fiberboard container conforming to PPP-B-636, Type CF or SF, Domestic Class, Variety SW, Grade 275. The bagged pairs of the drawers or undershirts, as applicable, shall be placed in the container as follows:

a. In two equal tiers.

b. The long dimension of the folded garment shall be parallel to the end panels of the container.

c. Every other garment shall be in a reverse position to the one on the top of it.

d. A fiberboard divider shall be placed between the tiers.

The top and bottom of the contents of each shipping container shall be completely covered with a sheet of 30 pound minimum basis weight kraft paper conforming to UU-P-268, Grade B. Shipping containers of extra small drawers or extra small undershirts shall contain a paper spacer (fabricated by crumpling a 24 x 48 inch minimum sheet of 30 pound minimum basis weight kraft paper). The spacer shall be centrally placed atop the garments. The body joint and the top and bottom flaps shall be firmly glued together as specified in PPP-B-636. The fiberboard container shall not contain any metal fastenings or stitches. All the seams and joints shall be sealed with 3 inch minimum wide tape conforming to PPP-T-45, Type III, Grade A, B, or C. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636. Toward the end of the contract or when there are less than the specified amount for the same size per container, mixed sizes may be packed withing the same shipping container (see 5.3.2).

5.2.3 Level C. The drawers or undershirts, as applicable, packaged as specified in 5.1, shall be packed within exterior type shipping containers in a manner that shall insure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

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5.3 Marking. In addition to any special marking required by the contract or order (see 6.2(g)), the exterior containers shall be marked in accordance with MIL-STD-129 and shall include the size of the drawers or undershirts, as applicable, and the date of manufacture (month and year).

5.3.1 Poylethylene bag. Prior to heat sealing, a white paper label, with legible and durable black letters and numerals, shall be placed inside each polyethylene bag. The label shall be placed so that it shall be capable of being read through the bag. The paper label shall contain the stock number, name of the item, size, and quantity.

5.3.2 Mixed sizes container. A white paper label, with legible black letters and numerals, approximately 5 by 4 inches, shall be securely attached to one end and one side of each container, containing mixed sizes of the drawers or undershirts, as applicable. The label shall contain the following information:

MIXED SIZES
Sizes and quantity of each size

6.NOTES

6.1 Intended use. The CWU-43/P drawers and the CWU-44/P undershirts covered by this specificiation are intended to be worn as the undergarments for cold weather and an anti-exposure protective assembly intended for continuous wear and protection of the wearer from exposure to cold water, wind, and spray as a result of emergency water immersion at sea.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Which item is desired, i.e., drawers or undershirts (see 1.1).

c. Size of the drawers or undershirts, as applicable, and quantity desired (see 1.2).

d. Whether first article inspection is waived (see 4. 3.1).

e. Name and address of the first article inspection laboratory (see 4.3.1).

f. Selection of applicable levels of packaging and packing (see 5.1 through 5.1.2 and 5.2 through 5.2.3).

g. Whether any special markings are required (see 5.3).

6.3 Patterns. The patterns, for the drawers or undershirts, as applicable, will be furnished by the contracting officer, to the contractor, for use in cutting working patterns (see 3.4.1).

6.3.1 Standard and guide samples. For access to standard guide samples, address the acquiring office issuing invitation for bids.

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6.4 First article. When a first article inspection is required, the item should be a first production item. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previoulsy acquired or tested by the Government, and that bidders offering such products, who wish to rely on such poroduction or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Data. For the information of contractors and contracting officer, any of the data, specified in the applicable documents, listed in Section 2 of ths specification, or referenced lower-tier documents, need not be prepared for the Government and shall not be furnished to the Government, unless specified in the contract or order. The data, to be furnished, shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NavWeps Form 4200/25 (Drawings, Lists, and Specifications Required) shall be attached, where applicable.

6.6 Supersession data. This specification includes the requirements of AS 3698 of 16 March 1973.

6.7 Subject term (Keyword) listing.

Drawers
Undershirts
Anti-Exposure
Flyers

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - AS
Air Force - 99

Review Activity:
Air Force - 82

User Activity:
Air Force - 45

Preparing activity
Navy - AS
(Project No. 8415-0674)

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TABLE I. Stitching characteristics.

Stitch type	Width (Inches)	Stitches per inch
101	-	12-16
301	-	8-10
402	7/32	12-16
407	8/32	14-18
503	5/32	12-16
504	5/32	14-18
605	9/32	14-18
606	8/32	12-16
606 modified	8/32	12-16

TABLE II. Drawers and undershirts Dimensions in inches.

Location	Sizes					
	X-Small	Small	Medium	Large	Extra Large	Tolerances
Drawers						
a. Waist	10	12	14	16	18	±1/2
b. Back Rise	14-1/2	15	15-1/2	16	16-1/2	±1
c. Length	37-1/8	38-1/2	39-7/8	41-1/4	42-5/8	±1
d. Width of Calf	5-3/4	6-1/4	6-3/4	7-1/4	7-3/4	±1/2
e. Length of Fly Opening	4	4-1/2	4-1/2	5	5	±1/2
f. Length of the Cuff	4-3/4	4-3/4	4-3/4	4-3/4	4-3/4	±1/4
g. Width of the Cuff	4	4	4	4	4	±1/4
Undershirts						
h. Body Length	24-1/2	26	27-1/2	29-1/2	31	±1

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TABLE II. Drawers and undershirts Dimensions in inches - Continued.

Location	Sizes					
	X-Small	Small	Medium	Large	Extra Large	Tolerances
i. Body Width	13-1/2	15	16-1/2	18	19	±1
j. Armhole Opening	7-7/8	8-1/2	9-1/8	9-3/4	10-1/2	±1/2
k. Neck Opening	7-1/2	7-1/2	7-1/2	7-3/4	7-3/4	±1/2
l. Sleeve Inseam	21	21-1/2	22	22-1/2	23	±1
m. Shoulder Width	3-1/4	4	5	5-1/2	5-3/4	±1/2
n. Length of the Cuff	2-3/4	2-3/4	2-3/4	2-3/4	2-3/4	± 1/4
o. Width of the Cuff	2-3/4	2-3/4	2-3/4	3	3	±1/4
p. Length of the Mock Turtleneck	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2	±1/4

TABLE III. Quality conformance inspection.

Inspection	Paragraph	
	Requirement	Method
Group A		
Visual examination of the drawers or undershirts as applicable	3.3 through 3.4.3.2 3.4.3.2, 3.6 through 3.7 and Figures 1 through 12 as applicable	4.5.1.1 and Table IV
Dimensional check of the drawers or undershirts as applicable	3.5 and Table II	4.5.1.1, Table V, and 4.5.2 through 4.5.2.2 as applicable
Packaging	Section 5	4.5.1.2 and Table IV

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TABLE IV. Classification of defects for the visual examination of the drawers and undershirts.

Defect	Major	Minor
GENERAL		
a. Any hole, scissors or knife cut, tear, mend, burn or weakening defect such as smash, multiple float, loose slub, needle chew, or abraded area	X	
b. Any portion of any drawers or undershirt, as applicable, stiffened, hardened, or scorched by any process of manufacturing	X	
c. Any unsightly slub, spot, or stain; any discoloration of the cloth caused by the removal of any size marking ticket; any size marking ticket, excluding the metal fastening or sewn-on kind, not removed; any size marking identification stamping visible on the outer-side of any drawers or undershirt, as applicable; or any size marking adhesive mass adhering to the cloth		X
d. Any metal fastening device or sewn-on size marking ticket	X	
e. Three or more thread ends or sewn stitching end not trimmed to a maximum length of 3/8 inch or three or more thread scraps not removed.		X
f. Any run or dropped stitch	X	
g. Any snag, pull, slubby yarn, or knit too tight or loose (lacking of elasticity)		X
h. Wales in the body, collarette, or in any cuff, not in the vertical direction.	X	
i. Collarette cut with more than one closing seam or at any location than body shoulder seam.	X	
COMPONENTS AND ASSEMBLY		
a. Any component not as specified or any defect of a component or defect of assembly, not herein classified	1/	
b. Any component, component part, or required operation omitted, or any operation improperly performed, not herein classified	1/	
c. Any component part not cut in accordance with the patterns, not herein classified	1/	

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TABLE IV. Classification of defects for the visual examination of the drawers and undershirts.- Continued

Defect	Major	Minor
SEAMS AND STITCHING		
a. Accuracy of seaming:		
1. Any seam, hem, or attachment of any component twisted, puckered, or pleated; any part of the drawers or undershirt, as applicable, any waste cloth or component caught in an unrelated operation or stitching		X
2. Ends of Stitch Type 301 stitching:		
a. Any not securely backstitched	X	
b. Any secure but backstitched for a distance less than 1/4 inch or the backstitching is not overlapped on itself		X
3. Thread breaks:		
(a) Any not securely backstitched	X	
(b) Any secure but overstitched for a distance less than 1 inch		X
4. Width of any stitch, not as specified (see Table I)	1/	
5. Seam edge:		
(a) Any overlapped seam whose untrimmed edge is more than 1/4 but no more than 1/2 inch		X
(b) Any overlapped seam whose untrimmed edge is more then 1/2 inch	X	
b. Gage of stitching 2/:		
1. Any stitching irregular, uneven, or not gaged as specified, or not uniform		X

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TABLE IV. Classification of defects for the visual examination of the drawers and undershirts.- Continued

Defect	Major	Minor
SEAMS AND STITCHING (Continued)		
c. Open seams 3/:		
1. Any 1/4 inch or less		X
2. Any more than 1/4 inch	X	
d. Raw edges 4/:		
1. Any more than 1/4 inch but not more than 1/2 inch		X
2. Any more than 1/2 inch	X	
e. Seam and stitch type:		
1. Not specified seam or stitch type	1/	
f. Stitch tension:		
1. Any loose stitching 5/	1/	
2. Any tight stitching, resulting in the breakage of the stitches, when normal pull is applied 6/	X	
g. Stitches per inch 7/:		
1. Stitches more or less specified:		
(a) One or 2		X
(b) More than 2	X	
H. Bartacks:		
1. Any missing, loose, misplaced, or not serving the intended purpose		X
2. Length of bartack not as specified		X

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TABLE IV. Classification of defects for the visual examination of the drawers and undershirts.- Continued

Defect	Major	Minor
LABEL		
a. Size information missing, incorrect, incomplete, or illegible	X	
b. Other than the size information, incomplete, incorrect, or illegible		X
c. Manner of attaching does not permit the label to expand with the waistband of the drawers or with the cloth of the undershirt, as applicable 8/	X	
d. Misplaced, off center of the back by more than 1/2 inch		X
e. Any row of stitching not adequately securing the label to the underwear	X	

1/ The defect shall be classified as major, when it seriously affects the serviceability or appearance, otherwise it shall be classified as a minor defect.

2/ The gage of stitching shall be scored only, when the condition exists for more than 2 inches on any row of stitching.

3/ A seam shall be classified as open, when one or more stitches, joining a seam, are broken or when two or more continuous skipped stitches or run-offs occur. On double stitched seams, a seam shall be classified as open, when either one or both sides of the seam contain any of the aforementioned for a single stitched seam.

4/ Any edge that should be turned under, covered, or trimmed and is not, but is securely caught in the stitching, shall be classified as a raw edge. Any raw edge, that is not securely caught in the stitching, shall be classified as an open seam.

5/ A seam shall be classified as loose, when the top and bobbin threads are on the same surface.

6/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident, and not caused by the gathering of the material, the stitching shall be inspected for tight tension and breaking of stitches by exerting normal pull in the lengthwise direction of the stitching, by pulling the material taut to straighten out the seam. Puckering due to the gathering of the material shall be classified as an accuracy of seaming defect.

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7/ The stitches per inch defect shall be classified only, when the condition exists for more than 3 inches on any row of stitching.

8/ Grasp the waistband or the undershirt, as applicable, on both sides of the label and pull sharply, stretching the waistband or the cloth, as applicable, to determine if the stitches attaching the label break (see 3.6.1 and 3.6.2).

TABLE V. List of defects for the dimensions of the assembled drawers and undershirts.

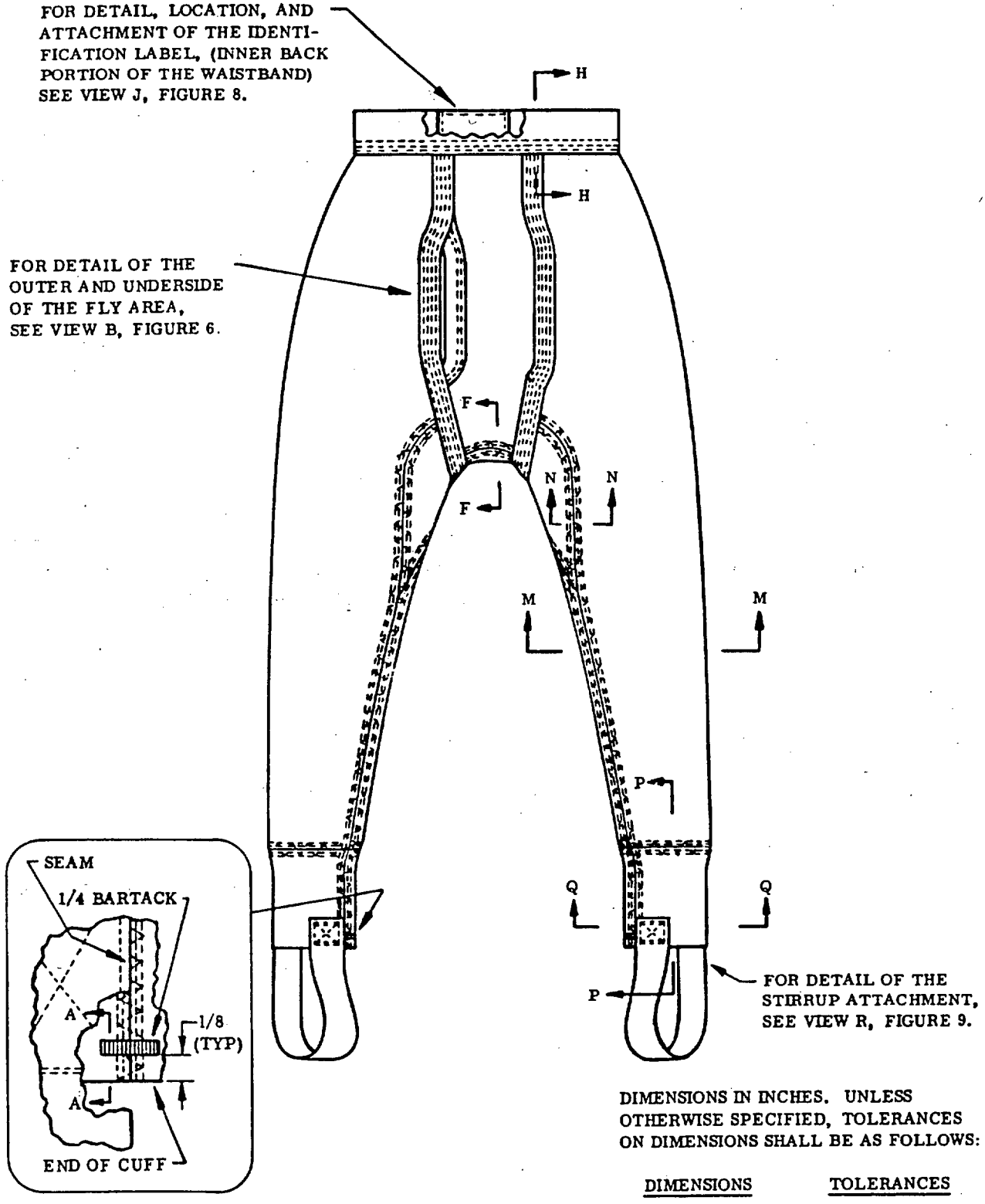
Finished measurements	Defect
DRAWERS a through g inclusive UNDERSHIRT h through p inclusive	Any measurement deviating from the measurements specified in 3.5 and Table II, and applicable tolerance, shall be enumerated as a dimensional defect.
EQUAL LENGTH Sleeves and legs, as applicable	The variation in the length between the left and right sleeve or leg, as applicable, shall be enumerated as a dimensional defect, when it exceeds 1/2 inch.

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TABLE VI. List of defects for packaging.

Item	Defect
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence, or method of application; size shown on the drawers or undershirts, as applicable, fiberboard container, or on the paper label not the same; information on the paper label in the polyethylene bag not readable through the film or no label in the film bag.
Packaging and packing materials	Any non-conforming component; any component or component part missing, damaged, or otherwise defective.
Workmanship	Inadequate aplication of the components such as incomplete closure of any exterior container, container flap, or loose strapping; bulging or distortion of any container; any shipping container contains any metal fastening or stitches; any polyethylene bag damaged, any vent hole missing, or any open or non-continuous heat sealed seam.
Exterior and interior weight or content	Gross or net weight exceeds the requirement; number per container more or less than specified; more than one size drawers or undershirts, as applicable, in the same container; any portion of any drawers or undershirt, as applicable, caught in any heat sealed seam; not individually packaged within the polyethylene bag; any drawers or undershirt, as applicable, distorted bent or deformed.

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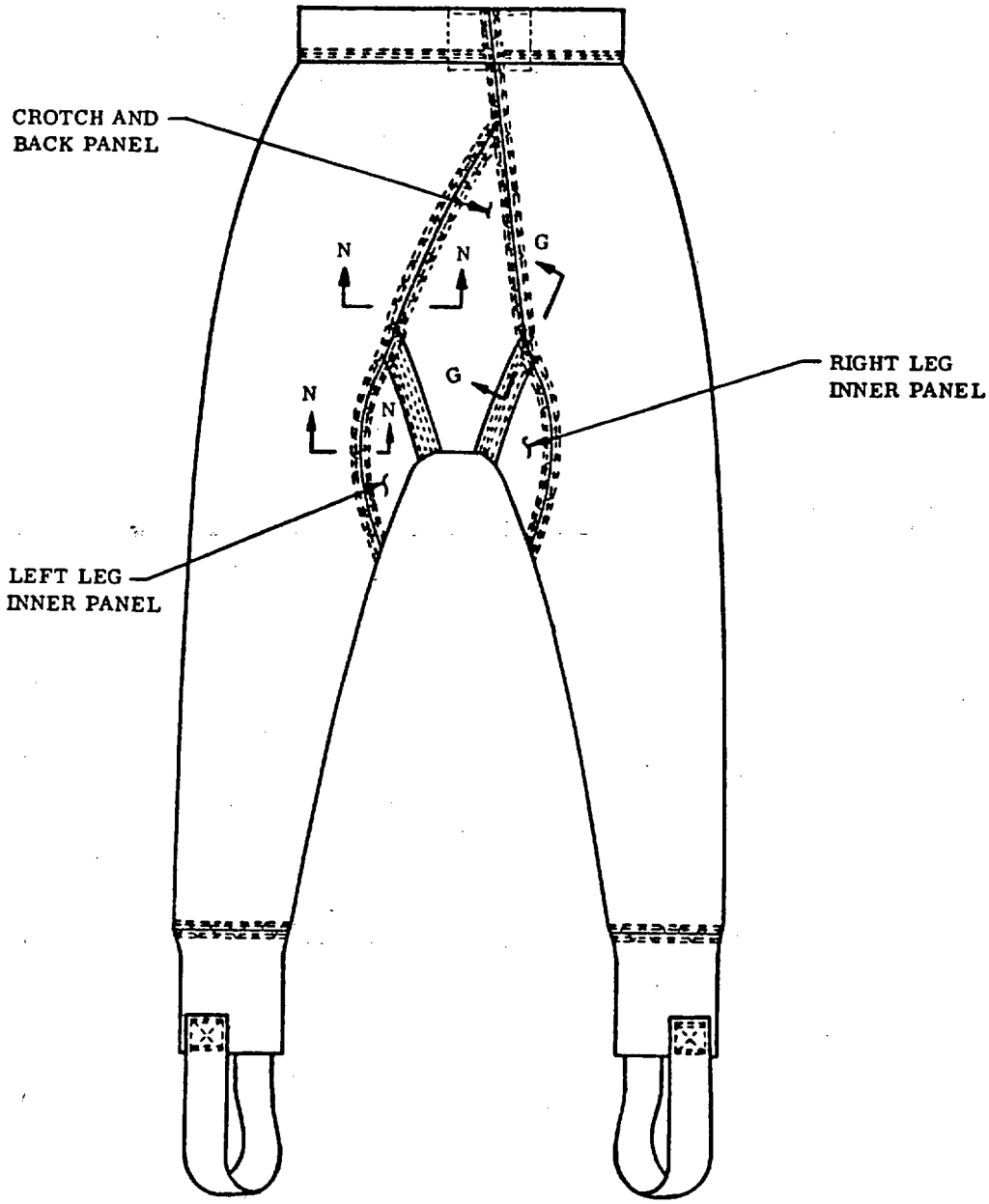
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES ON DIMENSIONS SHALL BE AS FOLLOWS:

DIMENSIONS	TOLERANCES
0 TO 1/2 INCLUSIVE	±1/32
OVER 1/2 TO 1 INCLUSIVE	±1/16
OVER 1 TO 6 INCLUSIVE	±1/8

FOR SECTION A-A, SEE FIGURE 4.
FOR SECTIONS F-F AND H-H, SEE FIGURE 7.
FOR SECTIONS M-M, N-N, P-P, AND Q-Q, SEE FIGURE 9.

FIGURE 1. Drawers - front

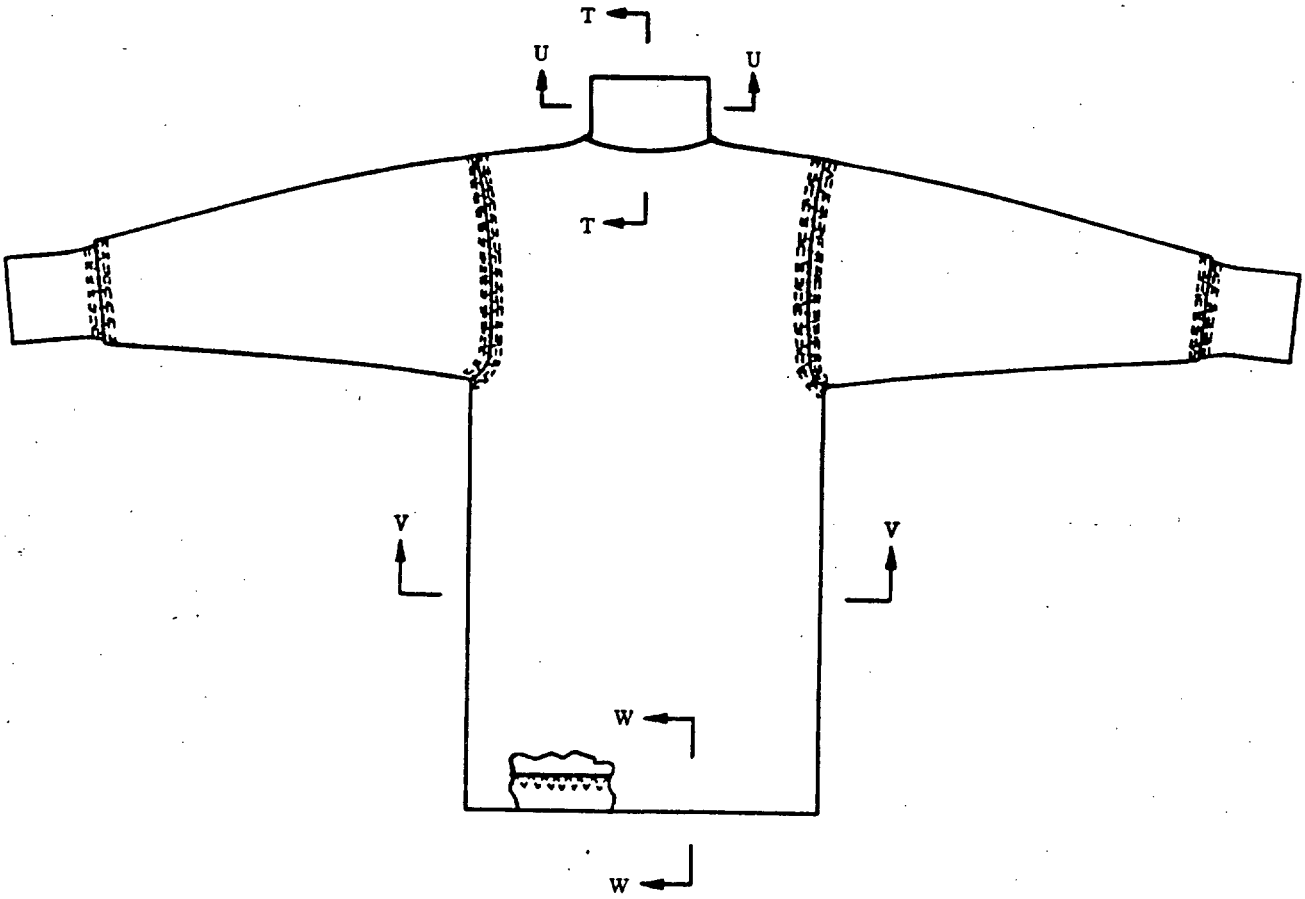
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FOR SECTION G-G, SEE FIGURE 7.
FOR SECTION N-N, SEE FIGURE 9.

FIGURE 2. Drawers - back

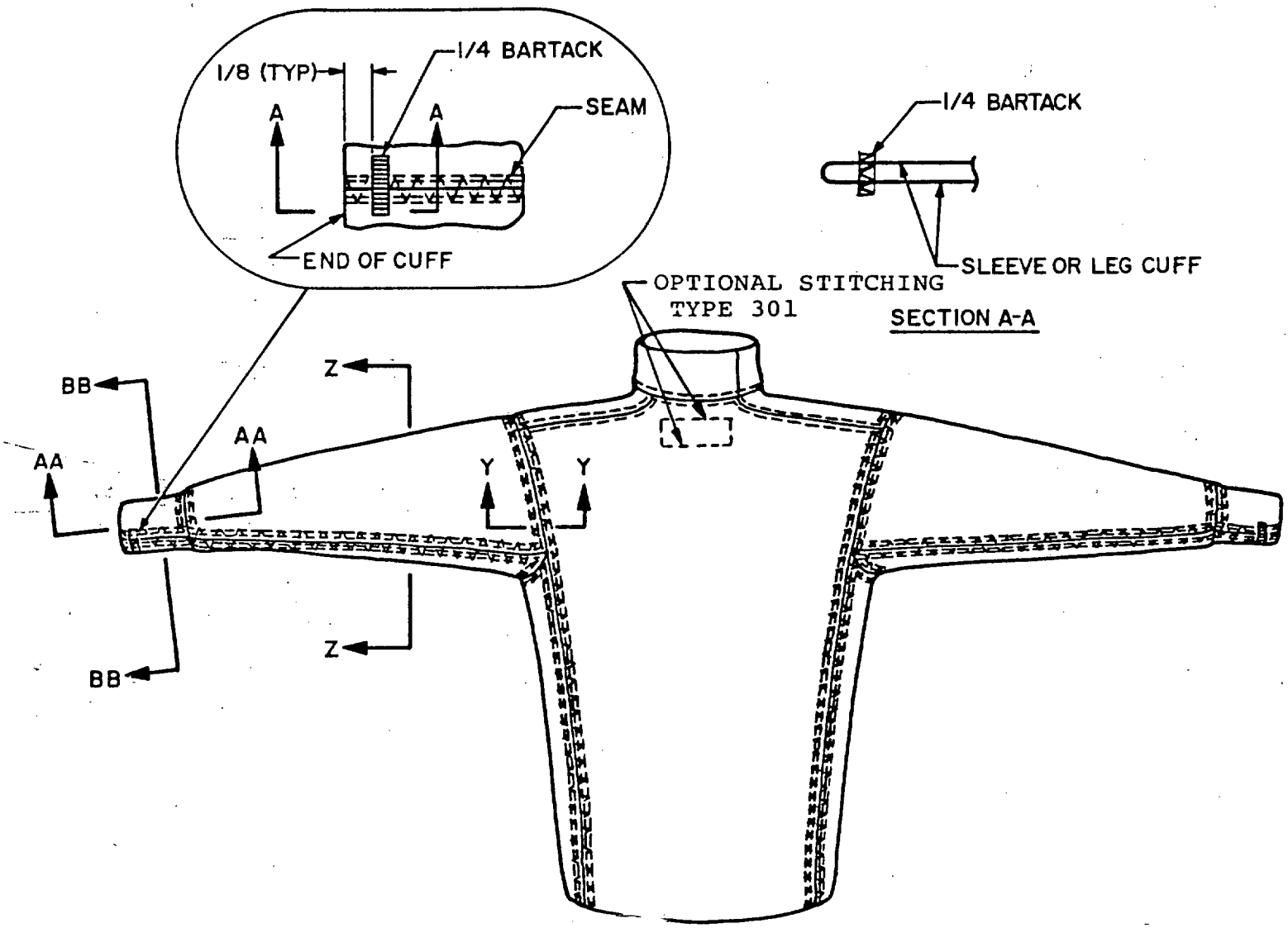
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FOR SECTIONS T-T, U-U, V-V, AND W-W, SEE FIGURE 10.

FIGURE 3. Undershirt - front

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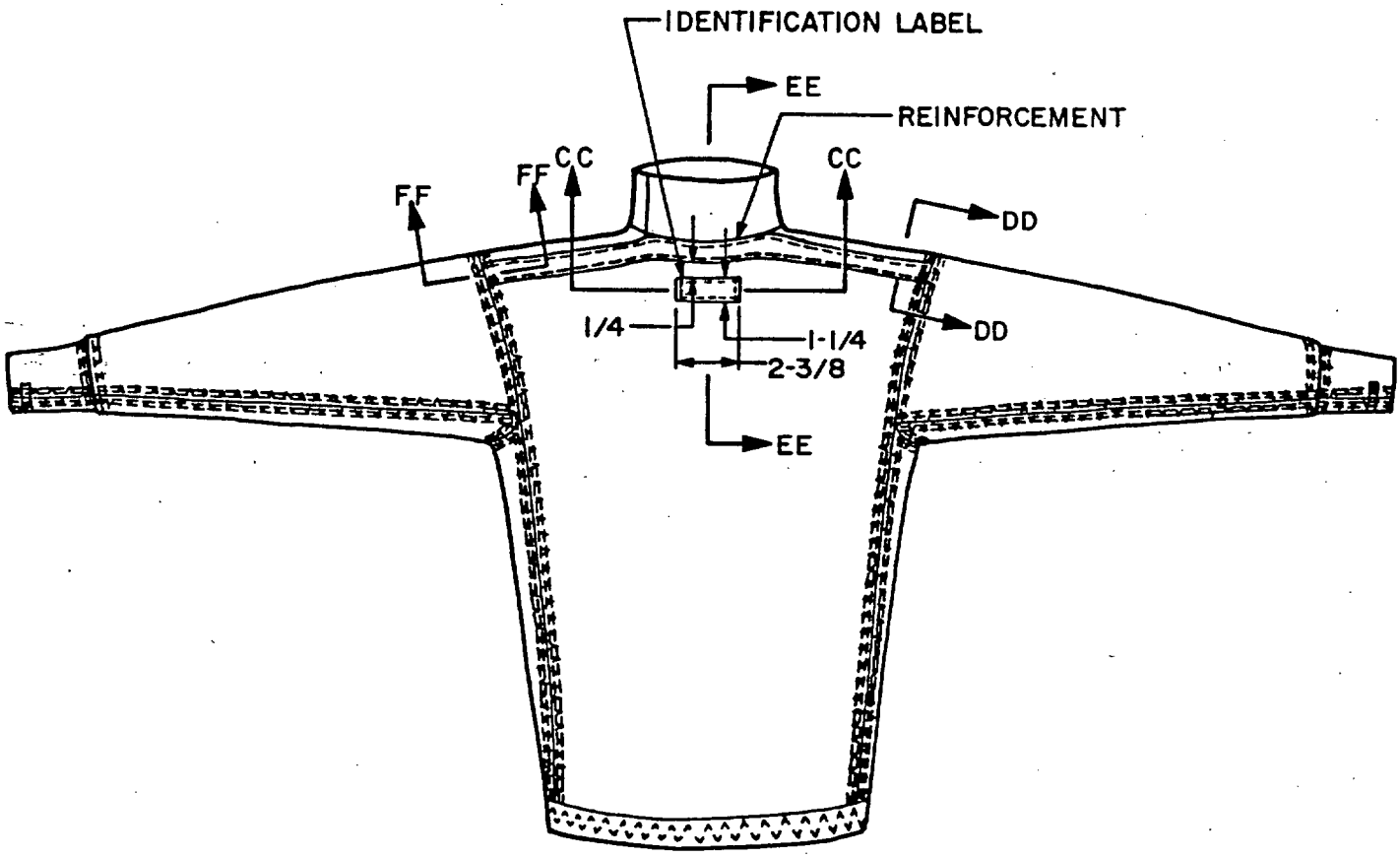


Dimensions in inches. Unless otherwise specified, tolerances shall be as specified in Figure 1.
Optional: Undershirt torso may be fabricated from tubular knit fabric.

For Sections Y-Y, Z-Z, AA-AA, and BB-BB, see Figure 12.

FIGURE 4. Undershirt - back

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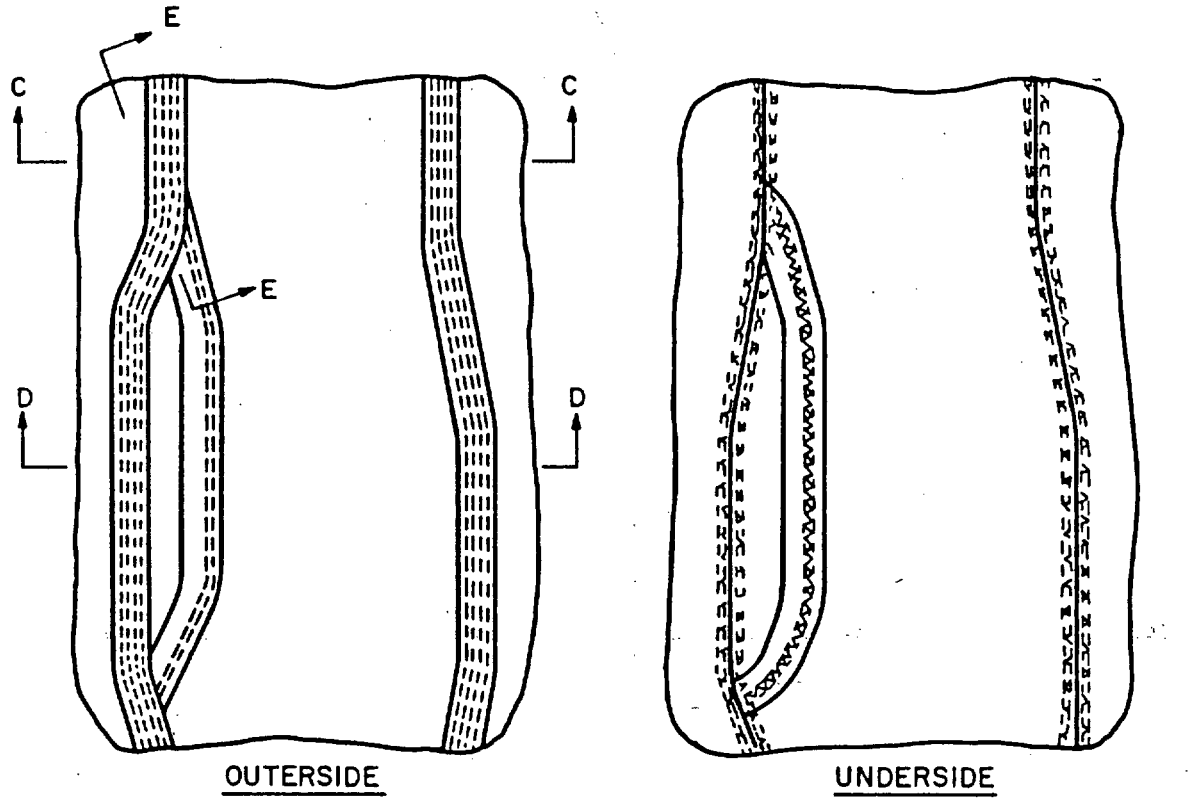


Dimensions in inches. Unless otherwise specified, tolerances shall be as specified in Figure 1.

For Section CC-CC, see Figure 11.
For Sections DD-DD, EE-EE, and FF-FF, see Figure 12.

FIGURE 5. Undershirt - back (innerside)

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Check the source to verify that this is the current version before use.

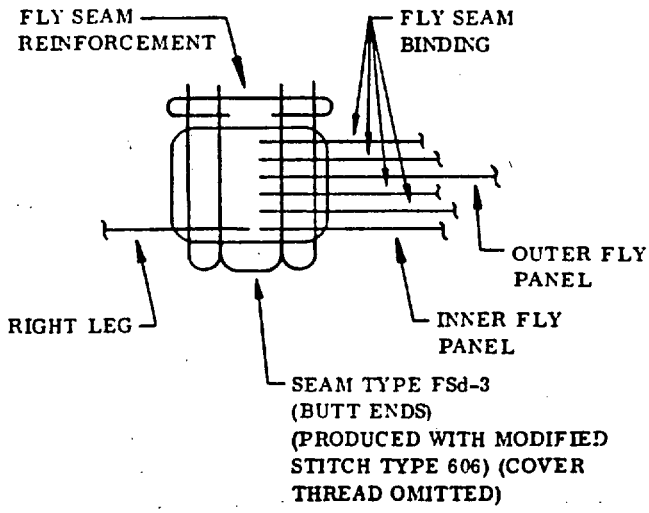


DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE AS SPECIFIED IN FIGURE 1.

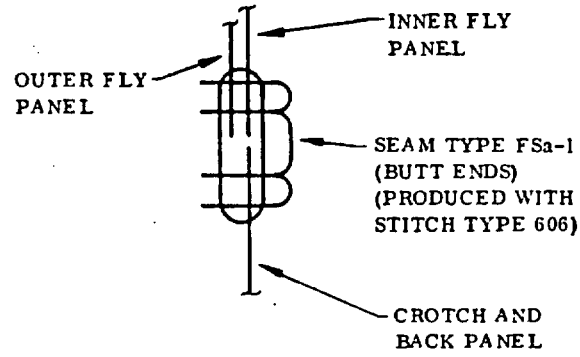
FOR SECTION E-E, SEE FIGURE 7.

FIGURE 6. Fly area sections

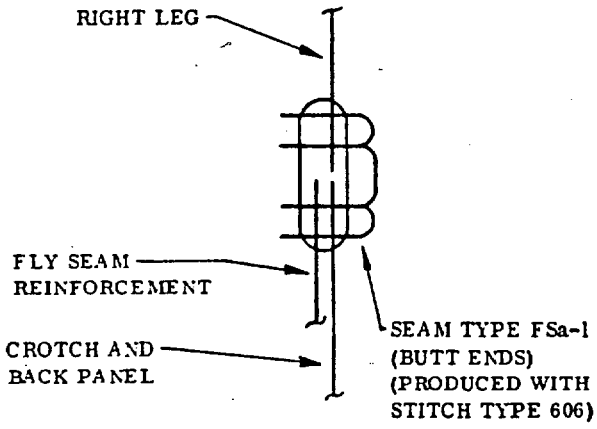
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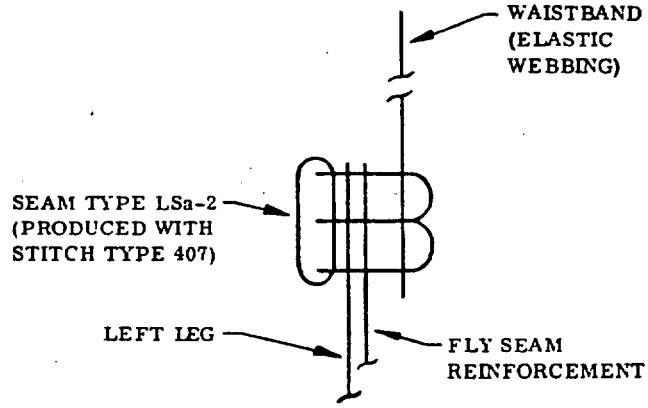
SECTION E-E



SECTION F-F



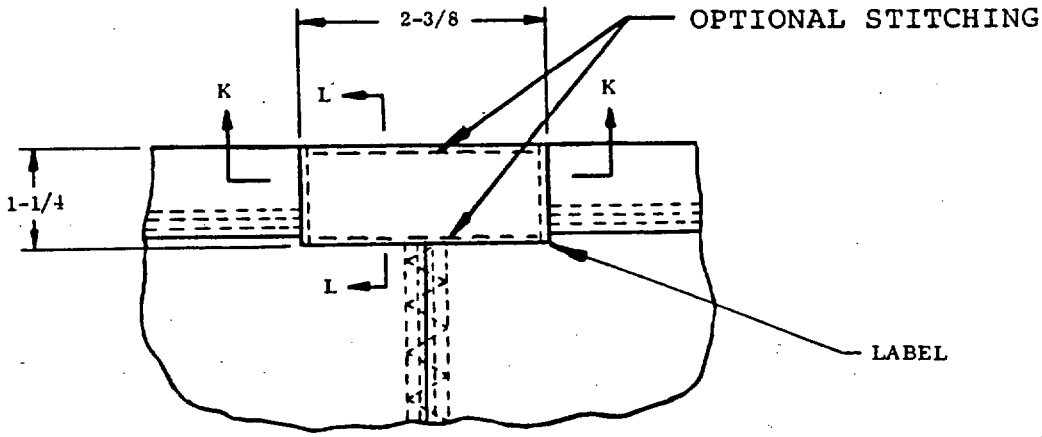
SECTION G-G



SECTION H-H

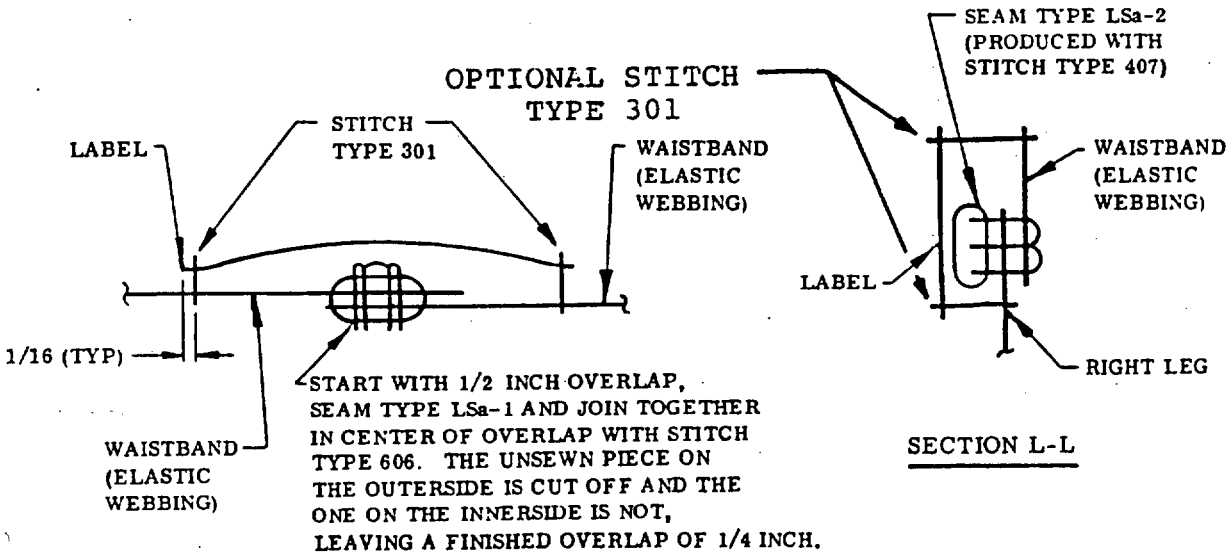
FIGURE 7. Fly area sections

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VIEW J

(INNERSIDE OF WAISTBAND)



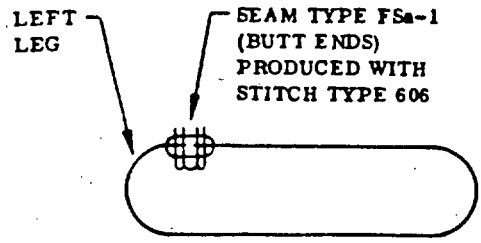
SECTION K-K

SECTION L-L

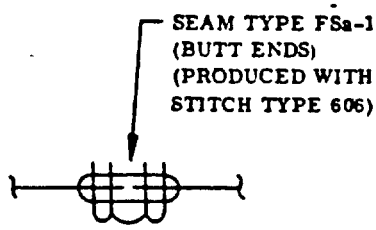
DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES SHALL BE AS
SPECIFIED IN FIGURE 1.

FIGURE 8. Waistband and label sections

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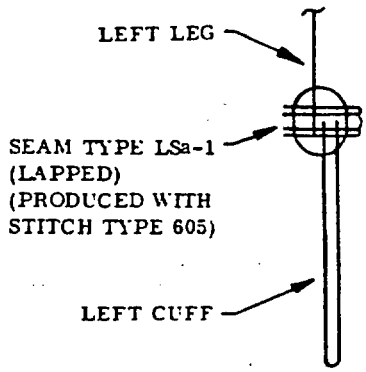


SECTION M-M

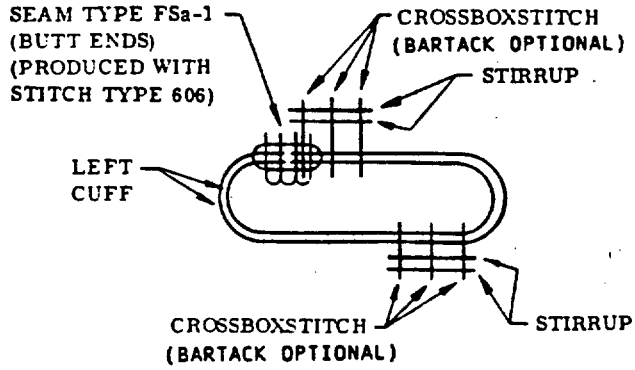


SECTION N-N

(TYPICAL FLAT SEAM CONSTRUCTION)

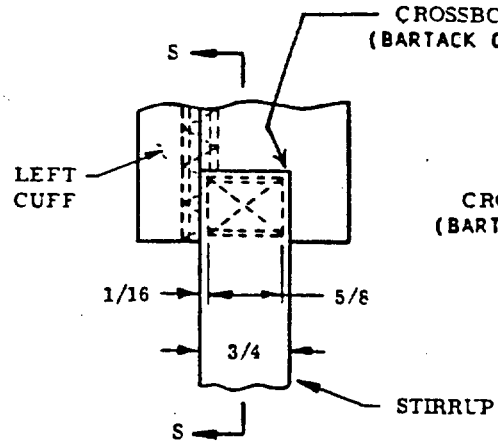


SECTION P-P

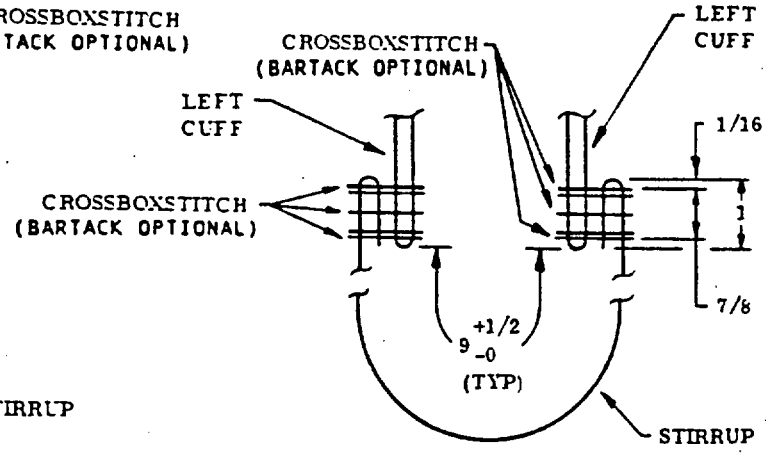


SECTION Q-Q

NOTE: SECTION P-P IS TO DESIGNATE A LAPPED SEAM WITH LEG FABRIC PLACED ON OUTSIDE SURFACE OF CUFF. LEG FABRIC OVERLAP IS NOT TO EXTEND BEYOND BOTTOM EDGE OF STITCH LINE.



VIEW R

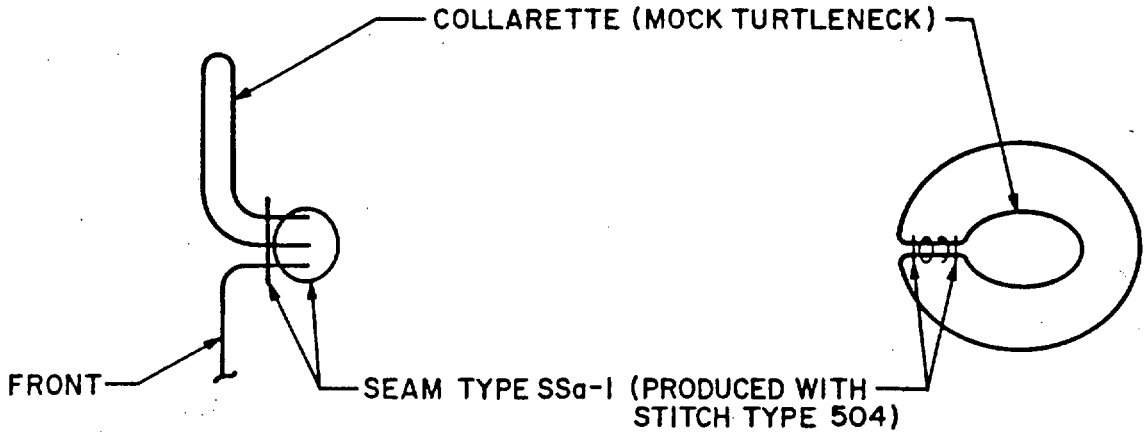


SECTION S-S

DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE AS SPECIFIED IN FIGURE 1.

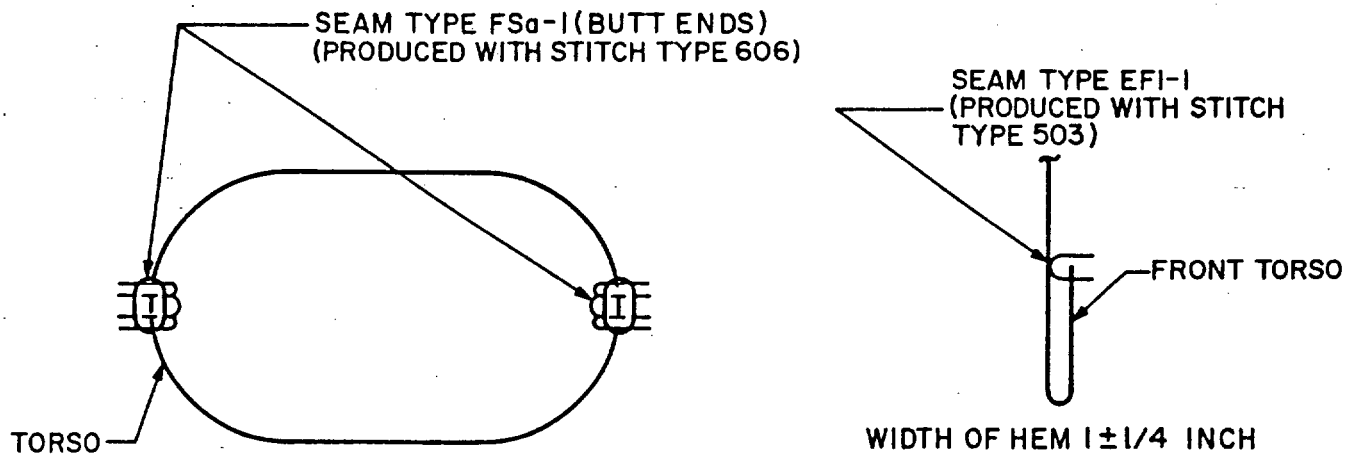
FIGURE 9. Drawers sections

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SECTION T-T

SECTION U-U



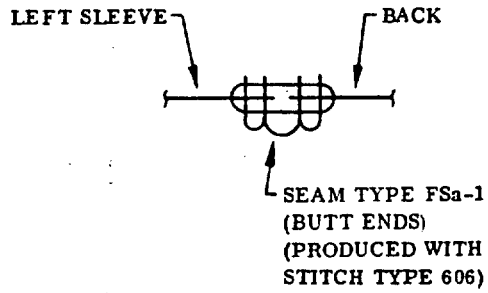
SECTION V-V

SECTION W-W

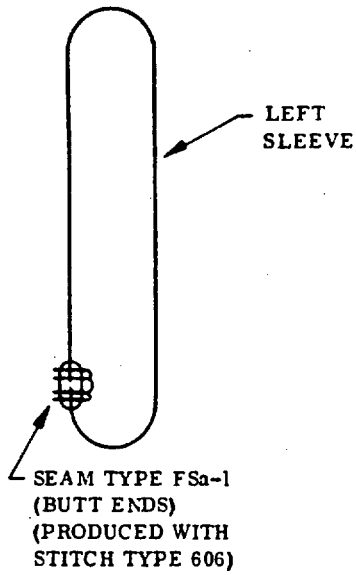
NOTE: SECTION V-V INDICATES SITE SEAMING IN THE UNDERSHIRT. THIS SEAMING IS OPTIONAL AND MAY BE ELIMINATED BY USING TUBULAR KNIT FABRIC.

FIGURE 10. Undershirt sections

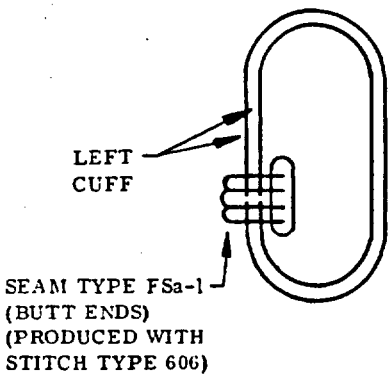
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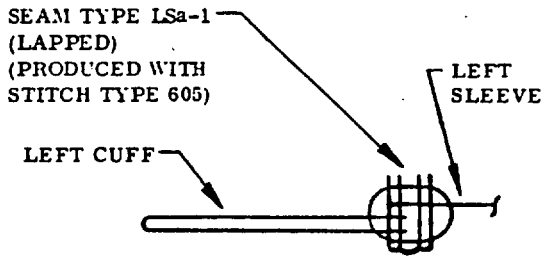
SECTION Y-Y



SECTION Z-Z

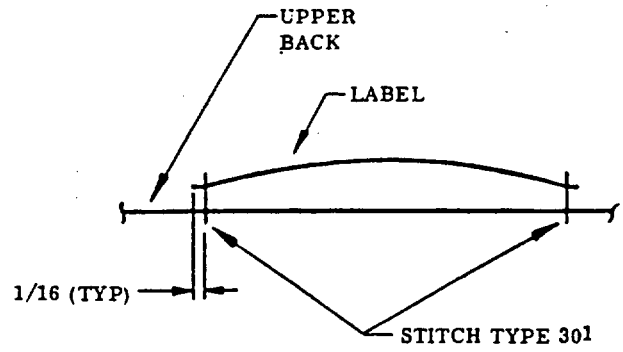


SECTION BB-BB



SECTION AA-AA

NOTE: SECTION AA-AA IS TO DESIGNATE A LAPPED SEAM WITH SLEEVE FABRIC PLACED ON OUTSIDE SURFACE OF CUFF. SLEEVE FABRIC OVERLAP IS NOT TO EXTEND BEYOND THE EDGE OF THE OUTER STITCH LINE.

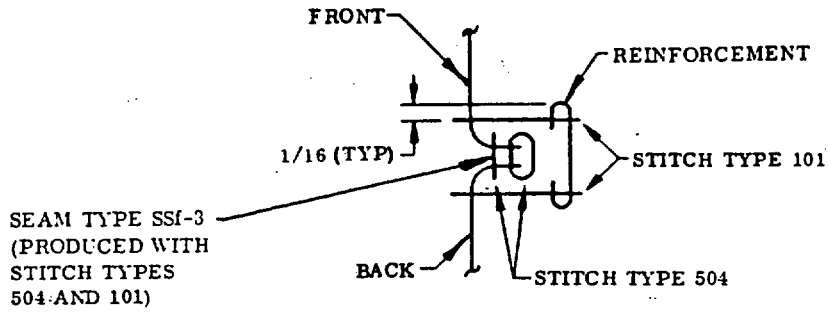


SECTION CC-CC

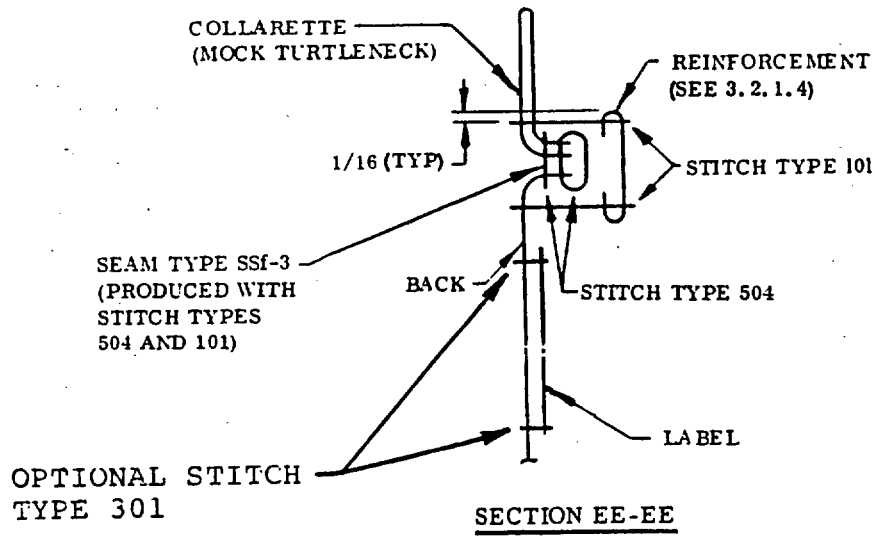
DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE AS SPECIFIED IN FIGURE 1.

FIGURE 11. Undershirt sections

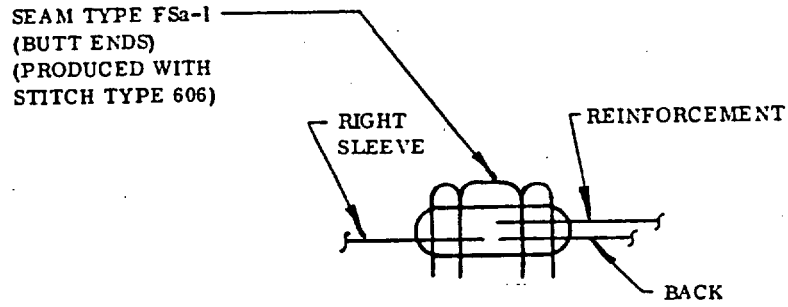
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SECTION DD-DD



SECTION EE-EE

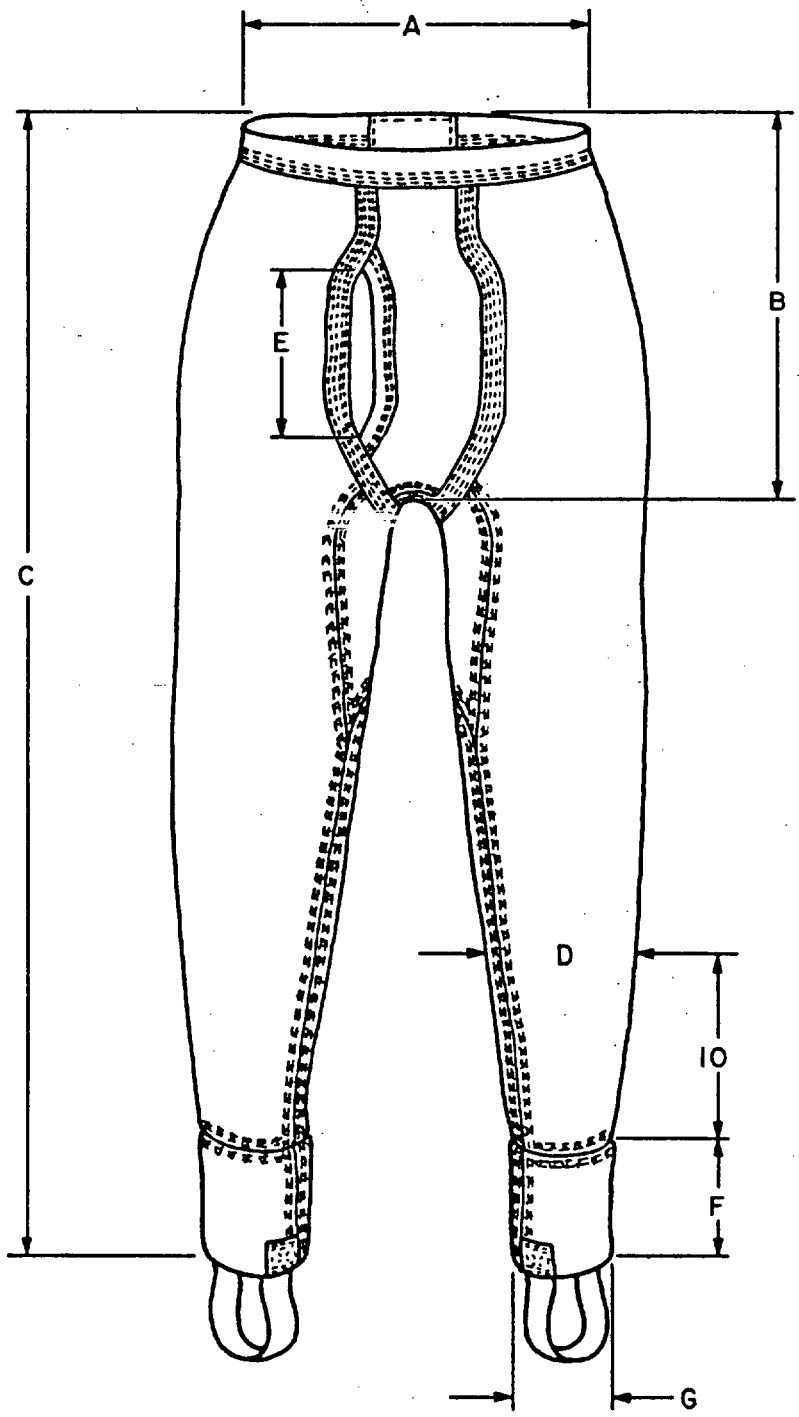


SECTION FF-FF

DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES SHALL BE AS
SPECIFIED IN FIGURE 1.

FIGURE 12. Undershirt sections

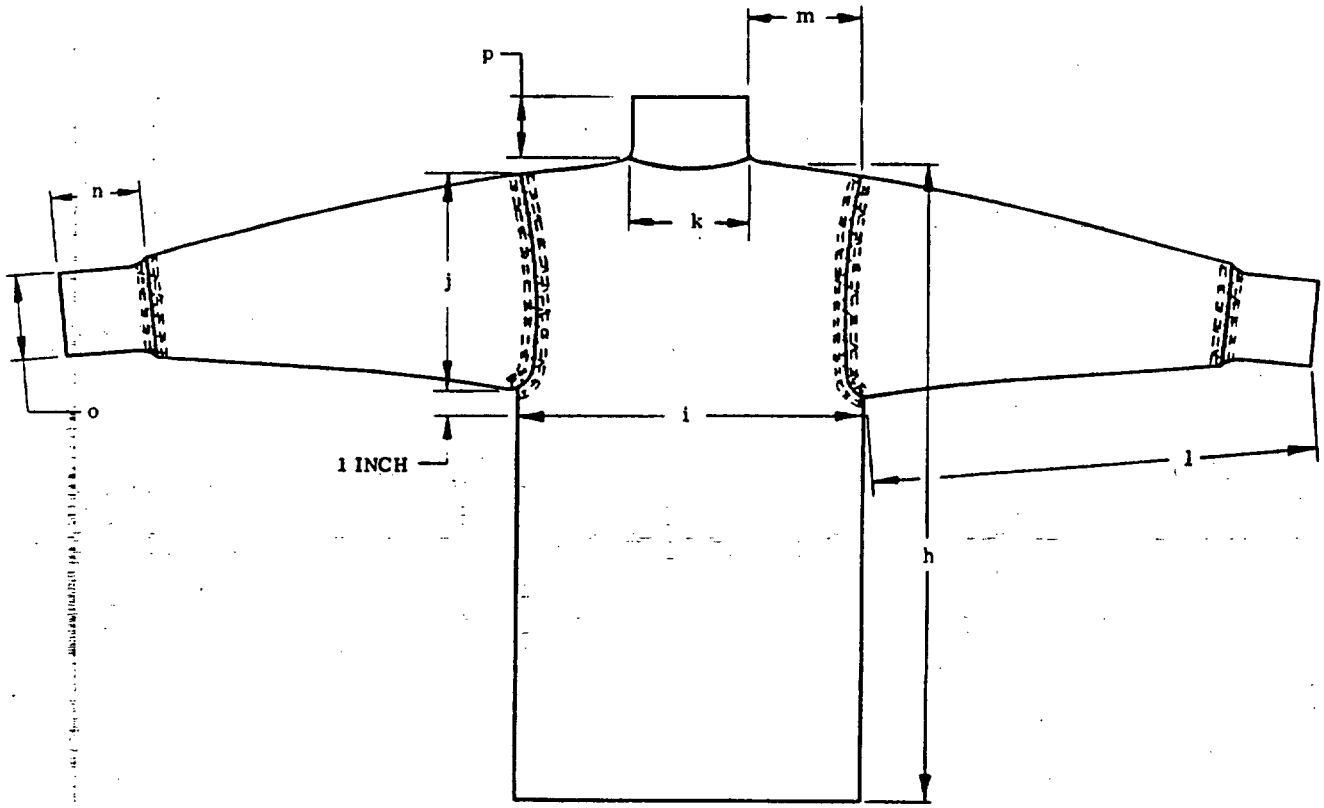
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Dimensions in inches.
For information see 4.5.2.1.

FIGURE 13. Location of drawers dimensions

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DIMENSIONS IN INCHES.
FOR INFORMATION SEE 4.5.2.2.

FIGURE 14. Location of undershirt dimensions

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-85040B	Drawers and Undershirts, Flyers, Anti-Exposure, Aramid, High Temperature Resistant, CWU-43/P & CWU-44/P

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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